

Working notes

A burst-aware extension of the Basic Model of Viral Replication

Consolidated BDC results, the renewal reformulation,
and an errata catalogue for Chapters 1, 3, 4, 5 and 6

Compiled August 3, 2026

Contents

1 Purpose and scope	4
2 Canonical notation for these notes	4
Part I — The single-cell BDC process, consolidated	5
3 Process and root algebra	5
3.1 The roots	5
3.2 The three fixation functions	6
3.3 Multiplicity of infection	7
4 Moments	7
4.1 Simplified closed forms	7
4.2 Late-time release and the mean burst size	7
5 Burst time, burst size, and the quasi-stationary distribution	8
5.1 Burst time	8
5.2 Burst size is exactly the quasi-stationary distribution	8
5.3 Size-biasing and the size of a late burst	9
5.4 Resolving the “curious circularity” of Chapter 4 §8	9
5.5 The burst-size law is now proved, not just conjectured	9
6 Second moment of the release, corrected	10
7 Summary table of single-cell results	11

Part II — The burst-aware BMVR	11
8 Why the Chapter 4 §9 attempt cannot be repaired as written	11
9 The correct object: a renewal (age-structured) BMVR	12
9.1 Kernels	12
9.2 The system	12
9.3 Relation to the literature (to be stated honestly in any write-up)	12
10 Effective BMVR parameters	13
10.1 Mean productive lifetime	13
10.2 Exact reduction to a constant-parameter BMVR	13
10.3 Closed forms for the transforms	14
11 Characteristic equation, R_0, and the growth-phase dependence of p	14
11.1 R_0 is untouched by bursting	14
11.2 p is not a constant of the pathogen	14
11.3 Identifiability (a negative result worth publishing)	15
12 The flooding advantage, answered exactly	15
12.1 Offspring distribution of one infected cell	15
12.2 Extinction probability in closed form	15
12.3 Comparison with constant budding at matched R_0	16
13 Multiplicity of infection	16
14 Two-type extension and the eclipse phase	17
15 Confrontation with single-cell data	17
Part III — Open tasks	18
16 Ordered by priority	18
Part IV — Errata and incomplete items	19
17 Chapter 1 — Introduction	19
18 Chapter 3 — BDC core	20

19 Chapter 4 — BDC additional and BMVR	21
20 Chapter 5 — BDC Two Type	23
21 Chapter 6 — Bursting and budding	23
22 Cross-cutting	24
23 Suggested paper outline	24
A Verification record	25

1 Purpose and scope

These notes do three things.

1. **Part I** consolidates the single-cell birth–death–catastrophe (BDC) results scattered across Chapters 3 and 4 into one verified, internally consistent set. Several formulas are simplified, several are corrected, and several results that the chapters have all the ingredients for but never state are stated here.
2. **Part II** sets out the extension of the Basic Model of Viral Replication (BMVR) that Chapter 4 §9 was reaching for. The attempt in §9 as written cannot be repaired — it is dimensionally inconsistent — but the object it was groping towards exists, is exactly solvable, and yields several results of publication value.
3. **Part IV** is a systematic catalogue of every error, gap, notational collision and structural problem found in the documents read. It is intended to be worked through and ticked off.

Everything asserted in Parts I and II has been checked, either algebraically or by independent numerical evaluation at $(\beta, \mu, \delta) = (1, 0.2, 0.05)$, $(1, 0, 0.1)$, $(1, 0.9, 0.1)$ and $(1, 0.5, 1/3)$. Where a result is *not* yet established, it is flagged as a Task in Part III rather than stated as a result.

2 Canonical notation for these notes

The thesis currently uses λ and β interchangeably for the birth rate, δ for both the catastrophe rate and the infected-cell death rate, K for both the second moment and the burst size, H and R for the same absorbing state, and I for both a probability and a cell count. These collisions become fatal the moment the intracellular and population models are joined, which is exactly what Part II does. The following choices are used throughout and should be adopted in the thesis.

Symbol	Meaning	Thesis usage
β	intracellular birth rate (per capita)	λ in Ch.3 §3, β in Ch.3 §4–§11
μ	intracellular death rate (per capita)	same
δ	catastrophe (burst) rate (per capita)	same; but also infected-cell death in Ch.1
X_t	intracellular count	same
R	absorbing rupture state	H in Ch.3, R in Ch.4/Ch.5
W_t	released (extracellular) count from one cell	W_t in Ch.3 §4, Y_t in Ch.3 §7
τ	burst time	same
$\mathcal{K}$	burst size (random)	$\mathcal{K}$ in Ch.3 §3, K in Ch.4 §7–§8
$I(t)$	$\mathbb{P}\{X_t \neq R\}$, no-burst probability	same
$D(t)$	$\mathbb{P}\{\text{cleared internally, no burst}\}$	<i>not named</i> ; equals $p_0(t)$ of Ch.4 §5
$\hat{I}(t)$	$\mathbb{P}\{X_t \notin \{0, R\}\}$, non-fixation	$\hat{I}$; formula wrong for $\mu > 0$
$J(t)$	$\mathbb{E}[X_t]$	same
$K(t)$	$\mathbb{E}[X_t^2]$	same
$V(t)$	$\mathbb{E}[W_t]$	same
θ	$\beta(a - b)$, the characteristic exponent	written out longhand everywhere
A, B	$a - 1$ and $1 - b$	not used; they simplify almost everything
T	target-cell count (BMVR)	S or T
γ	infection rate (BMVR)	β — collides with birth rate
$\mathcal{I}(t)$	infected-cell count (BMVR)	I — collides with $I(t)$
$\mathcal{V}(t)$	free-virion count (BMVR)	V — collides with $V(t)$
p	per-cell release rate (BMVR)	same
c	virion clearance rate (BMVR)	same
$d_{\mathcal{I}}$	infected-cell removal rate (BMVR)	δ — collides with catastrophe rate
r	population growth rate of the infection	not used

Part I — The single-cell BDC process, consolidated

3 Process and root algebra

One cell, one BDC process, $X_0 = 1$ unless stated. Per-capita rates: birth β , death μ , catastrophe δ . Catastrophe sends the process to the isolated absorbing state R and simultaneously sets $W_t = X_{\tau-}$.

3.1 The roots

$$\eta = \frac{\beta + \mu + \delta}{2\beta}, \quad a = \eta + \sqrt{\eta^2 - \frac{\mu}{\beta}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\beta}}. \quad (1)$$

Write

$$A := a - 1 > 0, \quad B := 1 - b > 0, \quad \theta := \beta(a - b) = \sqrt{(\beta + \mu + \delta)^2 - 4\beta\mu}. \quad (2)$$

Identities used constantly (all elementary, none stated in the thesis):

$$ab = \frac{\mu}{\beta}, \quad a + b = \frac{\beta + \mu + \delta}{\beta}, \quad (3)$$

$$AB = (a - 1)(1 - b) = \frac{\delta}{\beta}, \quad A + B = a - b. \quad (4)$$

Equation (4) is the one that does the work: it is why $b < 1 < a$ always (their product is positive and $\beta(1 - a)(1 - b) = -\delta < 0$), and it converts almost every messy expression in Chapter 3 into a short one. **It should be added to the reference table in Ch.3 §3.**

3.2 The three fixation functions

There are *three* absorbing-behaviour probabilities, not two. Chapter 3 defines two and gets the second one wrong.

$$I(t) = \mathbb{P}\{\text{no burst by } t\}, \quad (5)$$

$$D(t) = \mathbb{P}\{\text{internal extinction by } t, \text{ no burst}\}, \quad (6)$$

$$\hat{I}(t) = \mathbb{P}\{X_t \notin \{0, R\}\} = I(t) - D(t). \quad (7)$$

I and D both satisfy the same Riccati equation, because both events factorise over independent subtrees:

$$\frac{df}{dt} = \beta(f - a)(f - b) = \mu - (\beta + \mu + \delta)f + \beta f^2, \quad (8)$$

with $I(0) = 1$ and $D(0) = 0$. Writing $w := e^{\theta t} \geq 1$:

$$I(t) = \frac{aB + bAw}{B + Aw}, \quad D(t) = \frac{ab(w - 1)}{aw - b}, \quad (9)$$

$$\hat{I}(t) = \frac{(a - b)B}{B + Aw} + \frac{(a - b)b}{aw - b} = \frac{(a - b)^2 w}{(B + Aw)(aw - b)}. \quad (10)$$

Limits: $I(\infty) = b$, $D(\infty) = b$, $\hat{I}(\infty) = 0$, $\hat{I}(0) = 1$, $\hat{I}'(0) = -(\mu + \delta)$.

Correction (major). Chapter 3 §5 and §6.2 give $\hat{I}(t) = \hat{\eta} / (1 - (1 - \hat{\eta})e^{\hat{\eta}\beta t})$ with $\hat{\eta} = (\beta + \mu + \delta)/\beta$, derived from the claim $\hat{I}' = -(\beta + \mu + \delta)\hat{I} + \beta\hat{I}^2$. That branching term is wrong. On a birth event the two subtrees must satisfy *no burst in either, and not both extinct* — which is $I^2 - D^2$, not $\hat{I}^2$. The correct ODE is non-autonomous, $\hat{I}' = -(\beta + \mu + \delta)\hat{I} + \beta\hat{I}(\hat{I} + 2D)$, and the clean route is (7).

The error is invisible when $\mu = 0$ (then $D \equiv 0$ and $\hat{I} = I$), which is presumably why it survived. At $(\beta, \mu, \delta) = (1, 0.2, 0.05)$, $t = 1$: the correct value is $\hat{I} = 0.8087$, the thesis formula gives 0.6675.

Independent confirmation. Chapter 4 §5's state probabilities $p_n(t)$ sum (over $n \geq 1$) to exactly the right-hand side of (10) — verified algebraically. So Ch.4 already implies the correct $\hat{I}$ and contradicts Ch.3.

3.3 Multiplicity of infection

For $X_0 = k$ independent founders:

$$I_k(t) = I(t)^k, \quad D_k(t) = D(t)^k, \quad \boxed{\hat{I}_k(t) = I(t)^k - D(t)^k} \quad (11)$$

Chapter 3 §5 asserts $\hat{I}_k = \hat{I}^k$ and appends “(Is this correct?)”. It is not; the above is. The instinct was right.

4 Moments

4.1 Simplified closed forms

From $I' = -\delta J$ and (8):

$$J(t) = \mathbb{E}[X_t] = -\frac{1}{\delta} \frac{dI}{dt} = \frac{(a-b)^2 w}{(B+Aw)^2} = -\frac{\beta}{\delta} (I-a)(I-b), \quad (12)$$

$$K(t) = \mathbb{E}[X_t^2] = \left[1 + \frac{2\beta}{\delta} (1-I)\right] J = \frac{2\beta}{\delta} (\kappa - I) J, \quad \kappa = 1 + \frac{\delta}{2\beta}, \quad (13)$$

$$V(t) = \mathbb{E}[W_t] = (1-I) \left[1 + \frac{\beta}{\delta} (1-I)\right]. \quad (14)$$

Simplifications not in the thesis.

- Ch.3 §5 writes $J(t) = -\lambda(1-a)(1-b)(a-b)^2 e^{\beta(a-b)t} / [\delta(\dots)^2]$. Since $-\beta(1-a)(1-b) = \delta$ by (4), the β and δ cancel identically and J is just $(a-b)^2 w / (B+Aw)^2$, with $J(0) = 1$ manifest.
- Ch.3 gives $V = \frac{\beta-\mu}{\delta} (1-I) - J + 1$. This equals (14), which is a *quadratic in $(1-I)$ alone* — no J , no μ , no separate constant. Verified numerically at $(1, 0.2, 0.05)$, $t = 1$: both give 0.17843.
- Consequently $\delta K = V' = \frac{\beta}{\delta} \frac{d}{dt} (I - \kappa)^2$, i.e. $V = \frac{\beta}{\delta} [(I - \kappa)^2 - (1 - \kappa)^2]$, a useful alternative.

4.2 Late-time release and the mean burst size

$$\boxed{V_\infty = B \left(1 + \frac{\beta}{\delta} B\right) = \frac{a(1-b)}{a-1} = \frac{aB}{A}.} \quad (15)$$

The middle form is Ch.3's $\frac{\beta-\mu}{\delta}(1-b)+1$ rewritten; the right-hand form is new and much the most useful. Verified at $(1, 0.2, 0.05)$: both give 13.9857.

V_∞ is $\mathbb{E}[\mathcal{K}1\{\text{burst}\}]$ — it *includes* the realisations that cleared internally and released nothing. The conditional mean burst size is

$$\mathbb{E}[\mathcal{K} \mid \text{burst}] = \frac{V_\infty}{1-b} = \frac{a}{a-1} = \frac{\beta-\mu}{\delta} + \frac{1}{1-b}. \quad (16)$$

The last equality is Ch.3 §5's formula; the middle one is new and trivial by (15). Chapter 4 §8 flags this distinction with “(check)” — it is now checked, and both forms agree.

5 Burst time, burst size, and the quasi-stationary distribution

5.1 Burst time

$$\mathbb{P}\{\tau > t\} = I(t), \quad \varphi(t) := \text{density of } \tau = -I'(t) = \delta J(t), \quad \int_0^\infty \varphi = 1 - b. \quad (17)$$

This is defective: mass b sits at $\tau = \infty$. Chapter 4 §9 refers to “the known pdf of rupture times $\phi(t)$ ” without ever writing it down; it is $\delta J(t)$, and it has been available since $I' = -\delta J$ in Chapter 3 §7.

5.2 Burst size is exactly the quasi-stationary distribution

Chapter 4 §5 gives, for $n \geq 1$,

$$p_n(t) = \left(\frac{a-b}{b-aw}\right)^2 w P(t)^{n-1}, \quad P(t) := \frac{1-w}{b-aw}, \quad w = e^{\theta t}. \quad (18)$$

The n -dependence is entirely in P^{n-1} , so *at every time* t , conditional on $X_t \geq 1$, the load is geometric with ratio $P(t)$; and $P(t) \rightarrow 1/a$. Hence:

$$\text{QSD}(n) = \lim_{t \rightarrow \infty} \mathbb{P}\{X_t = n \mid X_t \notin \{0, R\}\} = \left(1 - \frac{1}{a}\right) a^{-(n-1)} = (a-1) a^{-n}, \quad n \geq 1. \quad (19)$$

$$\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\} = \frac{(\delta/\beta)a^{-k}}{1-b} = (a-1)a^{-k}, \quad k \geq 1. \quad (20)$$

New identity. *The burst-size distribution is exactly the quasi-stationary distribution.* Both are geometric on $\{1, 2, \dots\}$ with ratio $1/a$. This is not obvious — bursts are size-biased draws from the transient load distribution, integrated against the burst intensity — and it deserves a conceptual proof (Task 2). It is a striking, quotable exact result and neither chapter notices it.

Immediate consequences, none of which appear in the thesis:

$$\langle X \rangle_{\text{QS}} = \frac{a}{a-1} = \frac{a}{A}, \quad \langle X^2 \rangle_{\text{QS}} = \frac{a(a+1)}{(a-1)^2}, \quad \text{Var}_{\text{QS}}(X) = \frac{a}{(a-1)^2}. \quad (21)$$

Note $\langle X \rangle_{\text{QS}} = \mathbb{E}[\mathcal{K} \mid \text{burst}]$ and, when $\mu = 0$ only, both coincide with V_∞ — which is precisely the coincidence that Chapter 4 §6 observed and that Chapter 4 §9 then over-generalised. For $\mu > 0$ they differ: at $(1, 0.2, 0.05)$, $\langle X \rangle_{\text{QS}} = 17.232$ while $V_\infty = 13.986$.

5.3 Size-biasing and the size of a late burst

A cell bursts at rate δX_t , so the burst size given burst at time t is a *size-biased* draw from $p.(t)$:

$$\mathbb{P}\{\mathcal{K} = k \mid \tau = t\} = \frac{k p_k(t)}{J(t)}, \quad \mathbb{E}[\mathcal{K} \mid \tau = t] = \frac{K(t)}{J(t)}. \quad (22)$$

This is exactly the quantity Chapter 4 §9 calls $\mathbb{E}(R \mid \tau = t)$ and leaves undefined. Its late-time limit is

$$\lim_{t \rightarrow \infty} \mathbb{E}[\mathcal{K} \mid \tau = t] = \frac{2\beta}{\delta}(\kappa - b) = 1 + \frac{2\beta}{\delta}(1 - b), \quad (23)$$

so a very late burst is roughly twice the size of an average one — the signature of size-biasing a geometric.

5.4 Resolving the “curious circularity” of Chapter 4 §8

Chapter 4 §8 computes $\mathbb{E}[\mathcal{K} \mid \tau > t] = (\mathbb{E}[\mathcal{K}] - V(t))/I(t)$, finds that its $t \rightarrow \infty$ limit returns $\mathbb{E}[\mathcal{K}]$, and records that this is not understood.

The circularity is an artefact of using $\mathbb{E}[\mathcal{K}]$ to mean $\mathbb{E}[\mathcal{K} \mid \text{burst}]$ in the numerator while $I(t)$ in the denominator refers to the unconditioned event. With consistent definitions:

$$\mathbb{E}[\mathcal{K} \mathbf{1}\{\text{burst}\} \mid \tau > t] = \frac{V_\infty - V(t)}{I(t)} \xrightarrow{t \rightarrow \infty} \frac{V_\infty - V_\infty}{b} = 0, \quad (24)$$

which is correct and unmysterious: conditioning on “has not burst yet” converges to “never bursts”, on which the burst size is zero.

The genuinely interesting limit is obtained by conditioning on eventual burst,

$$\mathbb{E}[\mathcal{K} \mid \tau > t, \text{ burst eventually}] = \frac{V_\infty - V(t)}{I(t) - b} \xrightarrow{t \rightarrow \infty} \frac{K(t)}{J(t)} \Big|_{t \rightarrow \infty} = \frac{\langle X^2 \rangle_{\text{QS}}}{\langle X \rangle_{\text{QS}}}, \quad (25)$$

by l’Hôpital, since $V' = \delta K$ and $I' = -\delta J$. **This should replace the closing paragraph of Ch.4 §8.**

5.5 The burst-size law is now proved, not just conjectured

Chapter 4 §7 states $\mathbb{P}\{\mathcal{K} = k\} = \frac{\delta}{\beta} a^{-k}$, attributes the antiderivative to Wolfram Alpha, and says “The current goal is to verify this using Gillespie algorithm simulations”. No simulation is needed; the following is a proof.

With $P(t) = (1 - w)/(b - aw)$ and $w = e^{\theta t}$, direct differentiation gives

$$\frac{d}{dt} \left[\frac{P^k}{k\beta} \right] = \frac{1}{\beta} P^{k-1} P' = \left(\frac{a - b}{b - aw} \right)^2 w P^{k-1} = p_k(t), \quad (26)$$

using $P' = \beta(a - b)^2 w / (b - aw)^2$. Hence

$$\mathbb{P}\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k = \frac{\delta}{\beta} [P^k]_0^\infty = \frac{\delta}{\beta} \left(\frac{1}{a} \right)^k, \quad (27)$$

since $P(0) = 0$ and $P(\infty) = 1/a$. *Two* independent consistency checks now close the loop:

1. Normalisation: $\sum_{k \geq 1} \frac{\delta}{\beta} a^{-k} = \frac{\delta/\beta}{a-1} = 1 - b$ by (4), matching the eventual-burst probability.
2. First moment: $\sum_{k \geq 1} k \frac{\delta}{\beta} a^{-k} = \frac{\delta}{\beta} \frac{a}{(a-1)^2} = \frac{aB}{A} = V_\infty$, matching (15) — verified algebraically for general μ .

The second check is the important one: it independently validates the burst law, V_∞ , and the identity (4) simultaneously.

6 Second moment of the release, corrected

Chapter 3 §8 derives $\mathbb{E}[W_t^2]$ and makes two errors that propagate into $\text{Var}(W_t)$ and into Ch.3 §9.

Starting from $d\mathbb{E}[W_t^2]/dt = \delta\mathbb{E}[X_t^3]$ and $K' = 2(\beta - \mu)K + (\beta + \mu)J - \delta\mathbb{E}[X_t^3]$, eliminating $\mathbb{E}[X_t^3]$ and using $J = -\delta^{-1}I'$ (note the sign) and $K = \delta^{-1}V'$:

$$\frac{d\mathbb{E}[W_t^2]}{dt} = \frac{2(\beta - \mu)}{\delta} \frac{dV}{dt} - \frac{\beta + \mu}{\delta} \frac{dI}{dt} - \frac{dK}{dt}. \quad (28)$$

Integrating with the correct initial conditions $(I, J, K, V, \mathbb{E}[W^2])(0) = (1, 1, \mathbf{1}, 0, 0)$ — note $K(0) = \mathbb{E}[X_0^2] = 1$, *not* 0 as the thesis states —

$$\boxed{\mathbb{E}[W_t^2] = \frac{2(\beta - \mu)}{\delta} V - \frac{\beta + \mu}{\delta} I - K + \frac{\beta + \mu}{\delta} + 1,} \quad (29)$$

$$\text{Var}(W_t) = \mathbb{E}[W_t^2] - V^2. \quad (30)$$

Errors corrected. Chapter 3 §8 eq. (meanEY2) reads $\mathbb{E}[Y_t^2] = \frac{2(\beta - \mu)}{\delta} V + \frac{\beta + \mu}{\delta} I + K - \frac{\beta + \mu}{\delta}$: the signs on I and K are both wrong (traceable to the prose “using $J = \delta^{-1}dI/dt$ ”, which should be $J = -\delta^{-1}dI/dt$), and the constant $+1$ is missing because $K(0)$ was taken as 0.

Verification of (29). Take $\mu = 0$, $t \rightarrow \infty$: $I, K \rightarrow 0$, $V \rightarrow 1 + \beta/\delta$, giving $\mathbb{E}[W_\infty^2] = 2\beta^2/\delta^2 + 3\beta/\delta + 1$. Independently, $W_\infty = \mathcal{K} \sim \text{geometric on } \{1, 2, \dots\}$ with success probability $s = \delta/(\beta + \delta)$, whose second moment is $(2 - s)/s^2 = (2\beta + \delta)(\beta + \delta)/\delta^2 = 2\beta^2/\delta^2 + 3\beta/\delta + 1$. Exact agreement. The thesis formula gives a different number.

7 Summary table of single-cell results

Quantity	Closed form	Status
$I(t)$	$(aB + bAw)/(B + Aw)$	thesis, correct
$D(t)$	$ab(w - 1)/(aw - b)$	thesis (as p_0), not named
$\hat{I}(t)$	$(a - b)^2 w / [(B + Aw)(aw - b)]$	corrected here
$\hat{I}_k(t)$	$I^k - D^k$	corrected here
$J(t)$	$(a - b)^2 w / (B + Aw)^2$	simplified here
$K(t)$	$[1 + \frac{2\beta}{\delta}(1 - I)]J$	thesis, correct
$V(t)$	$(1 - I)[1 + \frac{\beta}{\delta}(1 - I)]$	simplified here
V_∞	$a(1 - b)/(a - 1)$	simplified here
$\mathbb{E}[W_t^2]$	eq. (29)	corrected here
$\varphi(t)$ (burst time)	$\delta J(t)$, mass $1 - b$	implicit only
$\mathbb{P}\{\mathcal{K} = k\}$	$(\delta/\beta)a^{-k}$	thesis; proved here
QSD	geometric($1/a$) on $\{1, 2, \dots\}$	new
$\langle X \rangle_{\text{QS}}$	$a/(a - 1)$	new
$\mathbb{E}[\mathcal{K} \mid \tau = t]$	$K(t)/J(t)$	new
$\int_0^\infty \hat{I}$	$\beta^{-1} \log(a/(a - 1))$	new , see §10.1

Part II — The burst-aware BMVR

8 Why the Chapter 4 §9 attempt cannot be repaired as written

Section 9 proposes replacing the constant p in $d\mathcal{V}/dt = p\mathcal{I} - c\mathcal{V}$ by $\langle X \rangle_{\text{QS}}$.

The obstruction is dimensional. p has units of particles per unit time. $\langle X \rangle_{\text{QS}}$ is a pure number. The proposed substitution therefore introduces a spurious factor of time.

Worse, by §10.1 below, when $\mu = 0$ one has $\langle X \rangle_{\text{QS}} = V_\infty$ exactly — so the proposal substitutes the *total lifetime output of one cell* in place of its *output rate*. This is why the section stalls, and no reweighting of a count by a probability density can fix it.

The two candidate integrals in §9 fail for related reasons:

- $p = \int \varphi(t) \mathbb{E}[\mathcal{K} \mid \tau = t] dt$ evaluates, using $\varphi = \delta J$ and $\mathbb{E}[\mathcal{K} \mid \tau = t] = K/J$, to $\delta \int K = V_\infty$ — again a count, not a rate.

- $p = \delta \int \varphi(t) \mathbb{E}[X_t^2 \mid \cdot] dt$ has the right dimensions *if* $\int \varphi dt$ is read as an averaging operator, but weights by the *burst-time* density. The correct weight is the *age distribution of currently infected cells*, which is a different measure entirely.

Section 9 also asserts $\mathcal{V}(t) = N_0 V(t)$ for N_0 initially infected cells. That is correct only if all N_0 cells are infected at the same instant. The section then notices the problem (“at any time, there will be infected cells that begin their infection dynamics at a number of different start times”) but does not act on it. Acting on it is the whole solution.

9 The correct object: a renewal (age-structured) BMVR

9.1 Kernels

Two kernels, both already in closed form from Part I:

cell-survival kernel: $\hat{I}(a) = \mathbb{P}\{\text{cell infected } a \text{ ago is still productively infected}\},$	(31)
release kernel: $g(a) = \delta K(a) = V'(a) = \text{expected release flux at cell-age } a.$	(32)

g is the correct object because a cell bursts at rate δX_t releasing X_t particles, so the expected release in $[a, a + da]$ is $\delta \mathbb{E}[X_t^2] da$. Chapter 4 §9 states this fact in its second paragraph and then does not use it.

9.2 The system

With T target cells (held constant, as in the simplified BMVR), infection rate γ , virion clearance c , and incidence $i(t) = \gamma T \mathcal{V}(t)$:

$i(t) = \gamma T \mathcal{V}(t),$	(33)
$\mathcal{I}(t) = \int_0^t i(t - \alpha) \hat{I}(\alpha) d\alpha,$	(34)
$\frac{d\mathcal{V}}{dt} = \int_0^t i(t - \alpha) \delta K(\alpha) d\alpha - c \mathcal{V}(t).$	(35)

(Initial-cohort terms are omitted for clarity; add $\mathcal{I}_0 \hat{I}(t)$ and $\mathcal{I}_0 \delta K(t)$ for cells present at $t = 0$.)

This replaces *both* BMVR terms: the constant release $p\mathcal{I}$ becomes a convolution, and the constant cell removal $d_{\mathcal{I}}\mathcal{I}$ becomes the age-structured form implicit in (34). Chapter 4 §9 only ever proposed to change the first.

9.3 Relation to the literature (to be stated honestly in any write-up)

The renewal/age-structured formulation itself is standard (McKendrick–von Foerster; Nelson–Gilchrist–Perelson-type age-structured within-host models; Euler–Lotka reduction from demography). **The novelty claim is not the framework.** It is that $\hat{I}(a)$ and $g(a)$ are here *derived in closed form from a mechanistic intracellular stochastic process* rather than posited phenomenologically, together with the consequences in §10–§12. Any submission must say this plainly in the introduction; claiming the framework will draw fire.

10 Effective BMVR parameters

10.1 Mean productive lifetime

$$\mathbb{E}[T_{\text{prod}}] = \int_0^\infty \hat{I}(t) dt = \frac{1}{\beta} \log\left(\frac{a}{a-1}\right) = \frac{1}{\beta} \log\langle X \rangle_{\text{QS}}. \quad (36)$$

Proof. Write $\hat{I} = (I - b) - (D - b)$ with $I - b = B(a - b)/(B + Aw)$ and $D - b = -b(a - b)/(aw - b)$; substitute $w = e^{\theta t}$, use $\theta = \beta(a - b)$, and integrate by partial fractions in w :

$$\int_0^\infty \hat{I} dt = \frac{1}{\beta} \left[\log \frac{a-b}{a-1} + \log \frac{a}{a-b} \right] = \frac{1}{\beta} \log \frac{a}{a-1}. \quad \square \quad (37)$$

When $\mu = 0$ this is $\beta^{-1} \log(1 + \beta/\delta)$. Checked numerically at $(1, 0.2, 0.05)$: formula 2.8468, trapezoidal integration 2.8541 at $h = 1$ (converging from above as expected). The relation to $\langle X \rangle_{\text{QS}}$ is a pleasant one and worth remarking on: the productive lifetime is the time it takes exponential growth at rate β to reach the quasi-stationary level.

10.2 Exact reduction to a constant-parameter BMVR

Substitute $i(t) = i_0 e^{rt}$ into (34)–(35) and compare with the BMVR equations $\mathcal{I}' = i - d_{\mathcal{I}} \mathcal{I}$, $\mathcal{V}' = p\mathcal{I} - c\mathcal{V}$. Writing $\tilde{f}(r) = \int_0^\infty e^{-r\alpha} f(\alpha) d\alpha$:

$$p_{\text{eff}}(r) = \frac{\delta \tilde{K}(r)}{\tilde{I}(r)}, \quad d_{\mathcal{I}, \text{eff}}(r) = \frac{1}{\tilde{I}(r)} - r. \quad (38)$$

This reduction is exact, not approximate, in the exponential phase. Substituting (38) into the BMVR eigenvalue relation $(r + d_{\mathcal{I}})(r + c) = \gamma T p$ returns $r + c = \gamma T \delta \tilde{K}(r)$, which is precisely the characteristic equation of the renewal system. So for any r there is a unique constant-parameter BMVR reproducing the renewal model's exponential solution exactly. It is *not* exact in transients.

Limits, both checked:

$$r = 0 : \quad p_{\text{eff}} = \frac{V_\infty}{\int_0^\infty \hat{I}} = \frac{\beta V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I}, \text{eff}} = \frac{\beta}{\log(a/(a-1))}, \quad (39)$$

$$r \rightarrow \infty : \quad p_{\text{eff}} \rightarrow \delta, \quad d_{\mathcal{I}, \text{eff}} \rightarrow \mu + \delta. \quad (40)$$

The $r = 0$ case uses $\int_0^\infty \delta K = V_\infty$ (immediate from $V' = \delta K$), which is the same answer as naive moment-matching — a useful consistency check. The $r \rightarrow \infty$ case is the correct young-cell limit: a newly infected cell holds one particle, bursts at rate δ , and releases one.

For $\mu = 0$ everything is elementary:

$$p_{\text{eff}}(0) = \frac{\beta(1 + \beta/\delta)}{\log(1 + \beta/\delta)}. \quad (41)$$

10.3 Closed forms for the transforms

With $s := r/\theta$ and $z := -B/A$:

$$\tilde{I}(r) = \frac{1}{\beta(s+1)} \left[\frac{B}{A} {}_2F_1\left(1, s+1; s+2; -\frac{B}{A}\right) + \frac{b}{a} {}_2F_1\left(1, s+1; s+2; \frac{b}{a}\right) \right], \quad (42)$$

$$\delta\tilde{K}(r) = \frac{\delta}{\beta} \left[\frac{{}_2F_1(1, s; s+2; z)}{A(s+1)} + \frac{2 {}_2F_1(2, s; s+3; z)}{A^2(s+1)(s+2)} \right]. \quad (43)$$

Both follow from the substitution $v = e^{-\theta t}$, which maps the integrals onto $\int_0^1 v^{s-1}(1-v)^n(A+Bv)^{-m}dv = A^{-m}B(s, n+1) {}_2F_1(m, s; s+n+1; z)$. Since $z < 0$ and $b/a < 1$, both series converge on the physical range. At $s = 0$ both reduce, via ${}_2F_1(1, 1; 2; z) = -\log(1-z)/z$, to $\int \hat{I} = \beta^{-1} \log(a/(a-1))$ and $\delta\tilde{K}(0) = V_\infty$ — both checks done.

Remark 10.1. The appearance of ${}_2F_1$ here and in Chapter 5’s two-type reduction is not a coincidence worth over-selling, but it does make the two chapters look like one project rather than two.

11 Characteristic equation, R_0 , and the growth-phase dependence of p

11.1 R_0 is untouched by bursting

The characteristic equation of (34)–(35) is

$$r + c = \gamma T \delta\tilde{K}(r), \quad \text{hence} \quad R_0 = \frac{\gamma T V_\infty}{c}. \quad (44)$$

Result of interest. At fixed total per-cell output V_∞ , replacing constant budding by bursting leaves R_0 *exactly unchanged*. Bursting alters the generation-time distribution — hence r , hence the extinction probability, hence the stochastic early phase — but not the threshold. This is the classic R_0 -versus- r distinction, and it connects the BMVR extension directly to the “perilous beginning” theme of Ch.3 §2 and to the flooding-advantage question of Ch.1 §6.

11.2 p is not a constant of the pathogen

Since p_{eff} depends on r and decreases from $p_{\text{eff}}(0)$ towards δ , the effective release rate is smaller during acute exponential growth (young cells dominate, loads are low) than at a steady state. At $(\beta, \mu, \delta) = (1, 0, 0.1)$: $p_{\text{eff}}(0) = 4.59$ against $p_{\text{eff}}(\infty) = 0.1$.

Prediction. BMVR fits to acute-phase data and to set-point data should yield systematically different p , in a direction and by an amount the theory specifies. Equivalently: the eclipse-like delay that is normally inserted by hand into within-host models emerges here from the intracellular dynamics, with no extra parameter.

11.3 Identifiability (a negative result worth publishing)

Three intracellular rates (β, μ, δ) map onto two BMVR parameters (p, d_I) at any given r . The map is therefore not invertible.

Consequence for Ch.4 §9's stated aim. The section's purpose is: “given someone ... has a good estimate for p ... they may then be able to access an approximation for the rates constituting the QS mean.” They cannot — not from p alone, and not from (p, d_I) either. A third observable is required. The natural candidates, in order of experimental accessibility:

1. the burst-size dispersion (the geometric ratio $1/a$ fixes one combination directly);
2. the fraction of infected cells that clear without bursting, which is b ;
3. the shape — not just the mean — of the release-time distribution $\varphi = \delta J$.

Any one of these plus (p, d_I) over-determines (β, μ, δ) , giving a consistency test. This should be stated as a result, not buried.

12 The flooding advantage, answered exactly

This is the strongest single result available and it addresses Ch.1 §6 and Ch.7's Komarova discussion directly.

12.1 Offspring distribution of one infected cell

Assume each released particle independently either infects (γT) or is cleared (c), so infects with probability $q = \gamma T / (\gamma T + c)$. With $y = 1 - q + qz$:

$$G_{\text{off}}(z) = b + \frac{\delta}{\beta} \frac{y}{a - y}, \quad (45)$$

which satisfies $G_{\text{off}}(1) = b + (1 - b) = 1$. Moments:

$$m = qV_{\infty} (= R_0 \text{ when } \gamma T \ll c), \quad \text{Var} = \frac{2q^2V_{\infty}}{a - 1} + qV_{\infty} - q^2V_{\infty}^2. \quad (46)$$

12.2 Extinction probability in closed form

$G_{\text{off}}(z) = z$ is a *quadratic* in y with $y = 1$ always a root. The other root gives

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1 - q + qb) - 1 + q}{q} = \frac{a - 1}{q} + 1 - a(1 - b). \quad (47)$$

12.3 Comparison with constant budding at matched R_0

Comparator: a BMVR cell with $\text{Exp}(d_{\mathcal{I}})$ lifetime producing at constant rate p , matched so that $p/d_{\mathcal{I}} = V_{\infty}$ and with the same q . Its offspring number is geometric with mean R_0 , so $z_{\text{ext}}^{\text{bud}} = 1/R_0$. Writing $L := a(1 - b) = a - \mu/\beta$, so that $V_{\infty} = L/(a - 1)$:

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{R_0} - 1 \right) = \left(a - \frac{\mu}{\beta} - 1 \right) \frac{1 - R_0}{R_0}. \quad (48)$$

Main result. For $R_0 > 1$, bursting strictly lowers the extinction probability relative to matched- R_0 budding if and only if $L > 1$, i.e.

$$a > 1 + \frac{\mu}{\beta} \iff \delta(\beta + \mu) > \beta\mu \iff \boxed{\frac{1}{\delta} < \frac{1}{\beta} + \frac{1}{\mu}} \quad (49)$$

When $L = 1$ the two extinction probabilities are *identical for every* q . For $R_0 \leq 1$ both models go extinct almost surely and the comparison is empty — note that (48) then changes sign, which is an artefact of both formulas running outside $[0, 1)$, not a reversal of the biology.

Interpretation: the flooding advantage is real, but it is bought against a risk. Bursting concentrates output into one batch, which lowers offspring variance relative to the geometric output of a budding cell; but it also carries the all-or-nothing hazard that the intracellular population dies out first, with probability b , and the cell releases nothing. The condition says the advantage survives exactly when the burst timescale $1/\delta$ is shorter than the harmonic-type combination of the birth and death timescales. When $\mu = 0$ the condition holds for all $\delta > 0$: bursting always wins.

Numerical verification. $(\beta, \mu, \delta) = (1, 0, 0.1)$, $q = 0.2$: $z^{\text{burst}} = 0.400$, $z^{\text{bud}} = 0.4545$, and (48) predicts -0.05455 — exact. $(1, 0.9, 0.1)$, $q = 0.9$: $z^{\text{burst}} = 0.9351$, $z^{\text{bud}} = 0.8442$ — the ordering reverses, as the condition requires. $(1, 0.5, 1/3)$ sits exactly at $L = 1$ and gives $z^{\text{burst}} = z^{\text{bud}}$ at both $q = 0.6$ and $q = 0.9$.

Remark 12.1. This deserves care in the write-up: the comparison holds the offspring *mean* fixed and is therefore a statement about branching structure only. It says nothing about nonlinear effects — immune saturation, local depletion — which is where Komarova’s own “all run out at once” intuition really lives. Both mechanisms should be named and separated.

13 Multiplicity of infection

For a cell founded by k particles, the kernels are

$$\hat{I}_k(a) = I(a)^k - D(a)^k, \quad g_k(a) = \delta[k K(a) + k(k - 1)J(a)^2], \quad (50)$$

the second from the variance of a sum of k i.i.d. copies. The release kernel is *superlinear* in k , so MOI matters even in the mean — a genuine prediction that the BMVR cannot express, and one that single-cell or low-MOI assays can test.

14 Two-type extension and the eclipse phase

Chapter 5 already provides the exact finite-time no-catastrophe probability $S(t)$ for the two-type process with type-specific rates (δ_1, δ_2) .

Observation not made in Chapter 5: GATE *is* an eclipse phase. The regime $\delta_1 = 0 < \delta_2$ means no burst is possible before conversion, so $S'(0) = 0$ — a zero initial hazard, followed by a rising one. That is precisely the eclipse-phase signature that within-host models normally insert by hand. In the two-type BDC it emerges mechanistically, with the eclipse duration set by ν .

This is the strongest reason to build the renewal model on the two-type kernel rather than the one-type kernel, and it makes Chapters 4 and 5 one story instead of two.

What is needed, and is *not* yet available:

- The release kernel is $g(a) = \delta_1 \mathbb{E}[X_a^2] + (\delta_1 + \delta_2) \mathbb{E}[X_a Y_a] + \delta_2 \mathbb{E}[Y_a^2]$ (all expectations over the killed process, i.e. including the indicator $\mathbf{1}_{\{\tau_c > a\}}$). These second moments are not in Chapter 5.
- The cell-survival kernel must be the two-type analogue of $\hat{I}$, i.e. “neither burst nor extinct”. Chapter 5 gives only S , the analogue of I , and explicitly notes that ordinary death does not stop its clock. The analogue of D is missing.

See Tasks 3 and 4.

15 Confrontation with single-cell data

The Hataye-type assay (one infected cell per well, viral counts hourly) measures the single-cell BDC directly, with no population model interposed. Predictions, all one-parameter or parameter-free given (β, μ, δ) :

1. A fraction b of wells should show *no release ever*.
2. Cumulative release per well is a step function: zero, then a single jump.
3. Jump times follow density $\delta J(t)/(1 - b)$.
4. Jump sizes follow geometric($1/a$) — and by the identity of §4.2, this is the same law as the quasi-stationary load.
5. Between-well variance is $\text{Var}(W_\infty)$ from (29), which the deterministic BMVR cannot produce at all.

Honest caveat that must be in the paper. HIV buds; it does not lyse in the all-or-nothing way the BDC assumes. Predictions (2) and (3) are therefore too strong for HIV as literally stated, and observed “intermittency” is not the same as a single terminal burst. Either (i) frame the one-type BDC as the bursting endpoint of a spectrum and fit an intermediate release model, which is exactly the budding–bursting spectrum of Chapter 6, or (ii) apply it to a genuinely

lytic system (*Y. pestis*, *F. tularensis*, lytic phage) where the assumption is right, and use HIV only as motivation. Option (ii) is safer; option (i) is the better paper if Chapter 6 can supply the intermediate model.

Part III — Open tasks

16 Ordered by priority

Task 1 (Literature check — do this first). Establish whether Carruthers et al. (*F. tularensis*) already couple an intracellular BDC to a between-cell model. If they do, the novelty claim must shift entirely onto the closed-form kernels, $p_{\text{eff}}(r)$, and (48). Also check: Pearson–Krapivsky–Perelson on early stochastic infection with burst versus budding; Nelson–Gilchrist–Perelson age-structured within-host; Gilchrist–Coombs nested models; the eclipse-phase fitting literature; Heffernan–Wahl on burst size. Confirm the BMVR attribution — Ch.1 cites McLean 1993, but Nowak–Bangham 1996 and Perelson et al. 1996 are the usual sources.

Task 2 (Prove the QSD/burst-size identity). Find a conceptual proof that $\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\}$ equals the quasi-stationary distribution. The algebra is done; a structural argument (size-biasing composed with the burst intensity, or a time-reversal/Yaglom argument) would make it quotable rather than coincidental.

Task 3 (Two-type killed second moments). Derive $\mathbb{E}[X_a^2 \mathbf{1}_{\tau_c > a}]$, $\mathbb{E}[X_a Y_a \mathbf{1}]$, $\mathbb{E}[Y_a^2 \mathbf{1}]$ for the Chapter 5 process. The moment hierarchy does not close upwards, exactly as in the one-type case; the one-type escape was to obtain K by differentiating the Riccati equation. The two-type analogue should be to differentiate the backward system (51)–(G) below. This is the main technical obstacle to the two-type renewal model and should not be assumed easy.

Task 4 (Two-type non-fixation probability). Construct the analogue of $D(t)$ for the two-type process (extinction without catastrophe) and hence the two-type $\hat{I}$. Chapter 5 supplies only S and G .

Task 5 (Growth-phase dependence, quantified). Plot $p_{\text{eff}}(r)$ and $d_{\mathcal{I}, \text{eff}}(r)$ over the physiological range of r for representative (β, μ, δ) , and quantify the predicted discrepancy between acute-phase and set-point BMVR fits. This is the figure that will sell the paper.

Task 6 (Characterise the flooding boundary). Map the region $\delta(\beta + \mu) > \beta\mu$ against plausible parameter ranges for *Y. pestis* in macrophages, and determine whether real pathogens sit on the advantageous side. Combine with (48) to give the *size* of the advantage.

Task 7 (Population-level variance). Propagate $\text{Var}(W_t)$ through the branching structure to a population-level prediction. For the single-cell assay this is unnecessary (§9 of Part II), but for whole-host data it is the main stochastic signal the BMVR cannot reproduce.

Task 8 (Intermediate release models). If option (i) of the data section is taken, define a release model interpolating between pure budding and pure bursting — e.g. partial release of a fraction of the load at catastrophe, or a per-capita export rate alongside the catastrophe rate — and check whether the kernel remains closed-form. This is the natural link to Chapter 6.

Task 9 (Nonlinear intracellular growth). A referee will object that linear birth lets the conditional load grow without bound. Prepare the defence (the burst hazard is proportional to load and

so regulates it; the QSD result is precisely the statement that the conditional load stabilises at $a/(a-1)$), and check numerically how far a logistic intracellular birth rate moves $g(a)$ and $\hat{I}(a)$.

For reference, the Chapter 5 backward system to be differentiated in Task 3:

$$\frac{dS}{dt} = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \quad (51)$$

$$\frac{dG}{dt} = \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2, \quad S(0) = G(0) = 1. \quad (52)$$

Part IV — Errata and incomplete items

Tags: **[ERROR]** mathematical error **[GAP]** unfinished **[NOTATION]** notation collision
[TYPO] typographical **[STRUCTURE]** structural.

17 Chapter 1 — Introduction

Folder: 1 Intro/document MAIN/

- 1.1 **[ERROR]** sections/06_hiv_and_the_basic_model_of_viral_replication.tex. The BMVR is written as $\dot{S} = \lambda - \beta SI$, $\dot{I} = \beta SI - \delta I$, $\dot{V} = pI - cV$. The infection term should be driven by V , not I : $\dot{I} = \beta SV - \delta I$. As written, V is entirely decoupled and plays no role in the dynamics — which undermines the whole section, since the point is that V is what the model gets wrong. Chapter 4 §9 uses the correct βTV form, so the two are inconsistent with each other.
- 1.2 **[ERROR]** Same file, simplified system: $\dot{I} = \beta TI - \delta I$, $\dot{V} = pI - cV$. Same problem; should be $\dot{I} = \beta TV - \delta I$.
- 1.3 **[NOTATION]** Same file. λ is the target-cell influx rate here but the intracellular *birth* rate in Ch.3. δ is the infected-cell death rate here but the *catastrophe* rate in Ch.3/4/5. β is the infection rate here but the birth rate in Ch.3. All three collide at exactly the point where the two models are supposed to meet. This is the single most damaging notational problem in the thesis.
- 1.4 **[NOTATION]** I and V denote cell and virion *counts* here, and *probabilities and expectations* in Ch.3/4. Ch.4 §9 writes both meanings within a page.
- 1.5 **[GAP]** Attribution: the BMVR is cited to `mclean1993balance`. Verify against Nowak & Bangham (1996) and Perelson et al. (1996), which are the conventional sources.
- 1.6 **[GAP]** Verify the Hataye et al. experimental details as stated (“2018”, “Los Alamos”, “32 small wells”, hourly sampling) against the published paper; the citation key is `hataye2019principles`, so the year in the text and the key disagree.
- 1.7 **[GAP]** sections/08_research_questions.tex lines 18–19. Research question 3 promises “a potential update to the standard BMVR”. As things stand the promise is not delivered — Ch.4 §9 states its own formula is incorrect. Either deliver (Part II above) or soften the claim.

- 1.8 **[NOTATION]** `main.tex` line 85 refers to “the basic model of viral replication (BMVR)” as a defined acronym before §6 introduces it. Check ordering.

18 Chapter 3 — BDC core

Folder: 3 BDC core/document MAIN/

- 3.1 **[ERROR]** (major) `sections/05_analytical_quantities.tex` and `sections/06_derivation_of_equations_and_results.tex`, §6.2. The formula $\hat{I}(t) = \hat{\eta}/(1 - (1 - \hat{\eta})e^{\hat{\eta}\beta t})$ and the ODE $\hat{I}' = -(\beta + \mu + \delta)\hat{I} + \beta\hat{I}^2$ are both wrong for $\mu > 0$. See Part I §3.2 for the correct $\hat{I} = I - D$ and for numerical evidence. Correct only when $\mu = 0$.
- 3.2 **[ERROR]** `sections/05_analytical_quantities.tex`. $\hat{I}_k(t) = [\hat{I}(t)]^k$ is wrong; correct is $I^k - D^k$. The text itself appends “(Is this correct?)”.
- 3.3 **[ERROR]** `sections/08_variance_and_standard_deviation_of.tex`, eq. (meanEY2). Signs on I and K are wrong and the constant $+1$ is missing. Two causes: the prose states $J = \delta^{-1}dI/dt$ where the sign should be negative; and the listed initial conditions give $K(0) = 0$ when in fact $K(0) = \mathbb{E}[X_0^2] = 1$. Corrected formula and an independent geometric-moment check in Part I §5.
- 3.4 **[ERROR]** Same file, eq. (VarY): inherits the above.
- 3.5 **[ERROR]** `sections/09_variable_dependencies_on.tex`. $K(I) = -\frac{2\beta^2}{\delta}(I - \kappa)(I - I_-)(I - I_+)$ contradicts `11_variable_dependencies_on.tex`, which gives $K(I) = \frac{2\beta^2}{\delta^2}(I - k)(I - I_+)(I - I_-)$. **The §11 version is correct**; §9 has both a wrong power of δ and a sign error. Derivation: $K = \frac{2\beta}{\delta}(\kappa - I)J$ with $J = -\frac{\beta}{\delta}(I - a)(I - b)$.
- 3.6 **[ERROR]** `sections/06_derivation_of_equations_and_results.tex`, eq. (ddIIfact): $\frac{dI}{dt} = \beta(I - I_+)(I - I_-)$ — the subscript is repeated; should be $(I - I_+)(I - I_-)$.
- 3.7 **[ERROR]** `sections/02_chapter_introduction.tex`: “The importance of stochasticity can not be understated” inverts the intended meaning. Should be “overstated” (or rephrase).
- 3.8 **[GAP]** `sections/05_analytical_quantities.tex`: the placeholder “PLOT $\mathbb{E}[\mathcal{K}]$ against δ for $\mu = 0$, $\lambda = 1$ ” is still in the source.
- 3.9 **[GAP]** `sections/05_analytical_quantities.tex`: $\mathbb{E}[\mathcal{K}]$ is never cross-checked against the geometric burst law of Ch.4 §7. They do agree; the check is in Part I §4.4 and should be added, since it validates several results at once.
- 3.10 **[STRUCTURE]** **Duplicated sections.** `08_variance_and_standard_deviation_of.tex` $\approx$ `10_variance_and_standard_deviation_of.tex`, and `09_variable_dependencies_on.tex` $\approx$ `11_variable_dependencies_on.tex`. All four are `\input` in `main.tex`, so the compiled PDF repeats the variance and I -dependency material with two conflicting versions of $K(I)$. Sections 10–11 are the earlier, shorter drafts. Decide which survives and delete the other pair.
- 3.11 **[STRUCTURE]** `sections/07_expected_number_of_units_j_t.tex` re-derives $J(t)$ and $V(t)$, already covered in §5. Third statement of the same material.

- 3.12 [NOTATION]** Birth rate is λ in the §3 reference table and in the $I(t)$, $J(t)$ formulas of §5, but β from §4 onwards. Two symbols, one rate, within one chapter.
- 3.13 [NOTATION]** The released count is W_t in §4 and the reference table, but Y_t in §7 and §8.
- 3.14 [NOTATION]** K is the second moment $\mathbb{E}[X_t^2]$ throughout, but $\mathcal{K}$ is the burst size in the §3 reference table, and Ch.4 §7–§8 use plain K for the burst size. Three uses of one letter.
- 3.15 [NOTATION]** The rupture state is H here and R in Ch.4 and Ch.5.
- 3.16 [NOTATION]** §5 defines $\hat{I}(t) = \mathbb{P}\{X_t \notin \{0, H\}\}$; §6.2 defines $\hat{I}(t) = \mathbb{P}\{X_t \neq 0\}$. These are different events. Fix to the former.
- 3.17 [TYPO]** `sections/03_reference_values.tex`: “ $p_k(t): \mathbb{P}\{X_t = i\}$ ” — index mismatch, and likewise for P_k and ϕ_k . Also ϕ_k is called a distribution with $\mathbb{P}\{\tau = i\}$, but τ is continuous: it is a density.
- 3.18 [TYPO]** `sections/04_process_definition.tex`: the transition list omits the conditioning on X_t and the $o(\Delta t)$ terms present in §6’s careful version. Also both figures carry `\label{BDRsims}` — duplicate label.
- 3.19 [TYPO]** `sections/02_chapter_introduction.tex`: “Brokwell” and “Brockwell” both used; “Jonathan Carrthurs” → Carruthers; “continuois”, “macrophaegs”, “mammels”, “populance”, “accross”, “sppured”, “Ricatti” → Riccati, “arives”, “encorporated”.
- 3.20 [STRUCTURE]** `sections/02_chapter_introduction.tex`: cross-references are stale. “This will be explored in chapter 3” and “explored further in chapter 4” point to material now in Chapters 6 and 7 of the current tree.

19 Chapter 4 — BDC additional and BMVR

Folder: 4 BDC additional and BMVR/document MAIN/

- 4.1 [ERROR]** `sections/09_potential_application_in_bmvr.tex`. The central proposal $p = \langle X \rangle_{\text{QS}}$ is dimensionally inconsistent: a count substituted for a rate. See Part II §6. The section admits at line 65 that “The above formula is incorrect” but attributes it to load-dependence of the burst rate, which is a second-order issue.
- 4.2 [GAP]** Same file, line 60: “(need to look up the formula for QS mean, but its something like $\lim J(t)/(1 - \hat{I}(t))$. I think there is an important point around whether $\mu = 0$ or not.)” — The formula is $\lim J/\hat{I} = a/(a - 1)$ (Part I §4.2); the μ point is real and is exactly that $\langle X \rangle_{\text{QS}} = V_\infty$ *only* when $\mu = 0$.
- 4.3 [GAP]** Same file: both candidate integrals are left unresolved. Neither works; see Part II §6 for why, and Part II §7–§8 for what replaces them.
- 4.4 [ERROR]** Same file: $\mathcal{V}(t) = N_0 V(t)$ holds only for synchronously infected cells. The section notices the asynchrony problem in the “Towards a correction” subsection but does not act on it.

- 4.5 **[GAP]** Same file: the closing subsection (“Iterating cycles”) describes the right object in words — a stationary regime of burst events with known time and size distributions — and stops. That object is the renewal kernel of Part II §7.
- 4.6 **[NOTATION]** Same file uses R for the rupture state while Ch.3 uses H ; and uses I for both the non-catastrophe probability and the infected-cell count in the same displayed equations.
- 4.7 **[TYPO]** Same file: the macro `\Vt` is defined in `main.tex` as `_{\text{total}}` V , which typesets a leading subscript with nothing to attach to. Almost certainly not intended.
- 4.8 **[GAP]** `sections/07_burst_size_distribution.tex`: “The current goal is to verify this using Gillespie algorithm simulations.” The result is correct and now has an analytic proof plus two consistency checks (Part I §4.4). Replace the note with the proof; a simulation figure is then a bonus, not a dependency.
- 4.9 **[TYPO]** Same file: the antiderivative is first written with exponent n and prefactor $(n\beta)^{-1}$, then restated with k ; and the same display appears twice.
- 4.10 **[GAP]** `sections/06_quasi_stationary_distribution.tex`: “There is a question of what form $F(t)$ takes, and how it differs between BDK and BDC.” It is determined by the section’s own decomposition: $F = 1 - \hat{I} = 1 - I + D$. Left open unnecessarily.
- 4.11 **[GAP]** Same file: the QSD is never identified. It is geometric($1/a$) on $\{1, 2, \dots\}$, for all μ , and follows immediately from the section’s own $p_n(t)$ (Part I §4.2). This is the chapter’s best unclaimed result.
- 4.12 **[TYPO]** Same file: the sentence “Ignoring, for now, the case of $i = 0$ & $i = R$, X_t has reached absorption, $\mathbb{P}\{X_t = i\} = 0$ ” is garbled; the intended statement is $\mathbb{P}\{X_t = i \mid \text{fixation}\} = 0$ for $i \notin \{0, R\}$.
- 4.13 **[TYPO]** Same file, figure caption: two different curves are both described as “yellow”.
- 4.14 **[ERROR]/[GAP]** `sections/08_further_considerations.tex`: the “curious circularity” is an artefact of using $\mathbb{E}[\mathcal{K}]$ for the conditional mean in the numerator and the unconditional event in the denominator. Correct limit is 0; the meaningful conditional limit is $\langle X^2 \rangle_{\text{QS}} / \langle X \rangle_{\text{QS}}$. See Part I §4.5. Also the “(check)” on $\mathbb{E}[\mathcal{K}] = V_{\text{inf}}$ is now discharged.
- 4.15 **[GAP]** `sections/03_probability_generating_function_pdes.tex`: in the catastrophe variant $G(z, t)$ is a *defective* PGF with $G(1, t) = I(t) \neq 1$. This is never stated and a reader will assume normalisation. The killing variant is proper.
- 4.16 **[TYPO]** Same file: “The probability generating function (PDF), is defined” — both the acronym and the spelling. Also “derivd”, “occurs”, “seperate”, “labeled” (inconsistent with British spelling elsewhere), “killng”.
- 4.17 **[STRUCTURE]** `sections/01_opening.tex` is empty but `\input`.
- 4.18 **[STRUCTURE]** The folder contains a complete stale duplicate of Chapter 3 at 4 BDC additional and BMVR/3 BDC core/. Delete or archive.

20 Chapter 5 — BDC Two Type

Folder: 5 BDC Two Type/Two_Type_Chapter/document MAIN/

- 5.1 **[GAP]** The two-type analogue of D (extinction without catastrophe) and hence of $\hat{I}$ is absent. §2 correctly notes that S and G “are no-catastrophe probabilities rather than population-survival probabilities”, which is exactly the I versus $\hat{I}$ distinction of Ch.3 — but the two chapters never cross-reference it, and Ch.5 provides only the I -analogue. Required for Part II. See Task 4.
- 5.2 **[GAP]** The killed second moments needed for a two-type release kernel are absent. See Task 3.
- 5.3 **[GAP]** GATE ($\delta_1 = 0 < \delta_2$, $S'(0) = 0$) is an eclipse phase. The chapter describes GATE purely as a biological extreme and does not make the connection to the within-host modelling literature, where the eclipse phase is a standing phenomenological patch. This is a strong, cheap addition.
- 5.4 **[STRUCTURE]** `sections/06_interpretation.tex` is a comment-only stub (“Content folded into `04_main_result.tex`”) but is still `\input`.
- 5.5 **[STRUCTURE]** The proof of the main theorem is commented out in `04_main_result.tex`. Deliberate or not, decide before submission.
- 5.6 **[STRUCTURE]** The chapter is written as a standalone paper (“The paper is organised around the derivation”; “A short conclusion and four technical appendices complete the paper”). Either commit to it as a paper and give the thesis a wrapper, or convert the self-references. Do not leave it ambiguous.
- 5.7 **[NOTATION]** λ_1, λ_2 are birth rates here; λ is the target-cell influx in Ch.1 and the birth rate in Ch.3 §3. μ_i are deaths, consistent. δ_i are catastrophe rates, consistent with Ch.3/4 but colliding with Ch.1’s infected-cell death rate.
- 5.8 **[NOTATION]** Absorbing state is R ; Ch.3 uses H .
- 5.9 **[GAP]** A figure caption embeds an external URL (`files.volvox.site/.../TT_bdc_genealogy_visualiser.html`). Confirm permanence and that a link is acceptable to the examiners/journal, or move it to a footnote with an archived copy.

21 Chapter 6 — Bursting and budding

- 6.1 **[GAP]** `sections/01_opening.tex` states that the BMVR is “consistent with an underlying stochastic model ... each infected cell releases virions according to a poisson process”. True, and the natural next sentence is that the resulting *offspring* distribution is geometric, not Poisson — which is precisely the comparator used in Part II §12. Worth adding; it makes Ch.6 and the BMVR extension continuous.
- 6.2 **[GAP]** The intermediate release model needed for Task 8 (partial release; the budding–bursting spectrum) is described qualitatively here. If it can be made explicit, it is the bridge between this chapter and the BMVR extension.

6.3 [TYPO] “ecclips phase” → eclipse; “wide speard” → widespread; “poisson” → Poisson; “non-cognisant”.

22 Cross-cutting

- X.1 [STRUCTURE]** Each chapter is a standalone `article` with its own preamble, macro block and `references.bib`. The macro blocks have already drifted apart (e.g. `\Vt`, `\Ic` exist only in Ch.3/4). Unify into a shared `preamble.tex` and a single `.bib` before the drift causes a real inconsistency.
- X.2 [STRUCTURE]** Chapter cross-references throughout Ch.3 §2 and Ch.1 §6 use the old numbering. Convert to `\Chapref` / `\cref` with real labels.
- X.3 [NOTATION]** Adopt the canonical notation table of §2 of these notes globally. The β/λ , δ/δ , I/I and V/V collisions must be resolved before Chapters 1 and 4 can be joined at all.
- X.4 [STRUCTURE]** Decide the identity of Chapter 5: thesis chapter, or paper with a thesis wrapper. The same decision then applies to the BMVR extension of Part II.

23 Suggested paper outline

Single paper, target *Journal of Theoretical Biology* or *Bulletin of Mathematical Biology*.

1. **Introduction.** The BMVR’s constant-release assumption; the single-cell evidence against it; what is and is not new here (be explicit — see Part II §7.3).
2. **The intracellular process.** BDC; the closed forms of Part I; the QSD and the burst-size law, with the identity between them.
3. **Renewal reformulation.** Equations (34)–(35); the kernels $\hat{I}$ and δK .
4. **Exact reduction.** $p_{\text{eff}}(r)$, $d_{\mathcal{I},\text{eff}}(r)$; the characteristic equation; R_0 unchanged; p depends on growth phase.
5. **Identifiability.** The negative result and the three rescuing observables.
6. **The flooding advantage.** Equation (48) and the condition $1/\delta < 1/\beta + 1/\mu$. This is the headline.
7. **Application.** Single-cell release data; the caveat of Part II §11.
8. **Discussion.** Limits of linear intracellular growth; the two-type/eclipse extension as future work (or fold it in if Tasks 3 and 4 are completed in time).

If the two-type tasks are completed, split into two papers: (i) the one-type theory as above; (ii) the two-type kernel with a mechanistic eclipse phase, building on Chapter 5.

A Verification record

Every result in Parts I and II was checked symbolically and then again numerically at $(\beta, \mu, \delta) \in \{(1, 0.2, 0.05), (1, 0, 0.1), (1, 0.9, 0.1), (1, 0.5, \frac{1}{3})\}$, and the stochastic claims were checked against an exact Gillespie simulation of the BDC (3×10^5 realisations) and a direct simulation of the cell-level branching process (6×10^4 realisations).

Claim	Check	Outcome
$ab = \mu/\beta, (a-1)(1-b) = \delta/\beta$	symbolic	exact
Three forms of V_∞ agree	symbolic + numeric	exact
$V(t)$ simplified = thesis form	numeric, 5 times	$< 10^{-10}$
$\hat{I} = (a-b)^2 w / [(B+Aw)(aw-b)] = I - D$	numeric	machine precision
$\sum_{n \geq 1} p_n(t) = \hat{I}(t)$	symbolic + numeric	exact
$\sum_k \mathbb{P}\{\mathcal{K} = k\} = 1 - b$	symbolic + numeric	exact
$\sum_k k \mathbb{P}\{\mathcal{K} = k\} = V_\infty$	symbolic + numeric	exact
QSD = burst-size law	symbolic	identical pmfs
$\langle X \rangle_{\text{QS}} = a/(a-1)$	numeric limit, $t = 40$	exact to 10^{-6}
$\int_0^\infty \hat{I} = \beta^{-1} \log \frac{a}{a-1}$	quadrature	$< 2 \times 10^{-3}$
$\mathbb{E}[W_t^2]$ corrected, eq. (29)	vs. exact geometric ($\mu = 0$)	exact, 4 values of δ
$z_{\text{ext}}^{\text{burst}}$ formula	branching simulation	within 10^{-3}
Flooding identity (48)	numeric, 6 parameter sets	exact

Simulation against theory, $(\beta, \mu, \delta) = (1, 0.2, 0.05)$

Quantity	Simulation	Theory
$\mathbb{P}\{\text{clears, no burst}\}$	0.18962	0.18839 ($= b$)
$I(1)$	0.92789	0.92730
$I(3)$	0.61773	0.61597
$I(8)$	0.20016	0.19891
$\mathbb{E}[W_\infty]$	13.957	13.986 ($= V_\infty$)
$\mathbb{E}[W_\infty^2]$	465.6	468.0 corrected eq. (29)
		428.1 thesis eq. (meanEY2) — <i>rejected</i>
$\mathbb{E}[\mathcal{K} \mid \text{burst}]$	17.223	17.232 ($= a/(a-1)$)
$\mathbb{P}\{\mathcal{K} = 1 \mid \text{burst}\}$	0.05753	0.05803 geometric($1/a$)
$\mathbb{P}\{\mathcal{K} = 5 \mid \text{burst}\}$	0.04615	0.04569
$\mathbb{P}\{\mathcal{K} = 25 \mid \text{burst}\}$	0.01379	0.01382

The $\mathbb{E}[W_\infty^2]$ row is the decisive one: the corrected formula sits within Monte-Carlo error of the

simulation, the thesis formula is 8.5% low — far outside it.

Extinction probabilities, branching simulation

(β, μ, δ)	q	formula	simulation	budding
$(1, 0, 0.1)$	0.2	0.4000	0.3986	0.4545
$(1, 0.9, 0.1)$	0.9	0.9351	0.9351	0.8442
$(1, 0.5, \frac{1}{3})$	0.6	0.8333	0.8328	0.8333

The three rows are, respectively: bursting wins ($L > 1$), bursting loses ($L < 1$), and the exact tie at $L = 1$ predicted by (48).

Numerical caution

$\hat{I}$ must be evaluated from the closed form (10), not as $I - D$. Both tend to b while their difference tends to 0, so the subtraction loses all significant figures for $\theta t \gtrsim 35$ — at $(\beta, \mu, \delta) = (1, 0.5, \frac{1}{3})$, $t = 40$, the subtraction returns 5×10^{-4} where the true ratio $J/\hat{I}$ is 3. Anything downstream that uses $\hat{I}$ at large t (notably $\int \hat{I}$ and $\tilde{\hat{I}}$) must use (10).